

NC DHHS Grant Finder: work product and process

UNC Solve-a-thon 2026 qualifier · Agency: NC Department of Health and Human Services.

This report shows how we got from the 1,662 Grants.gov listings to our ranked results. Along with it we submitted:

- **Source code (PDF):** every script, prompt and config file.
- **Results workbook (Excel):** every grant's numbers at every step.
- **AI-use statement (PDF):** tools, key prompts, and where the AI was wrong.

Our pipeline in one paragraph

Layer 0–1, plain Python rules (no AI): clean the data, remove 359 dead listings (closed, archived, paperwork notices), and score deadline and award size, leaving 1,303. **Layer 2, Jev:** a fast multiple-choice AI screens every grant with 4 questions and keeps 283, setting a grant aside only when it's 85–95% sure. **Layer 3, DeepSeek:** a large language model reviews the 283 and must back every claim with an exact quote, which our code verifies (282/283). **Layer 4, a published formula** (strategic fit 30%, can DHHS run it 25%, award 15%, deadline 15%, eligibility 15%) ranks them into 33 open grants, 116 forecasts and 134 research grants. **Layer 5, checks:** federal award history (USAspending.gov), hand labels, a keyword-search baseline, and an AI error log.

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Reproduce everything: `pip install -r requirements.txt`, then `python src/pipeline.py`. It reruns from saved model answers in about 11 seconds with no API key; there are 71 automated tests (`python -m pytest tests -q`).

Agency Profile: NC Department of Health and Human Services (NCDHHS)

Source: [NCDHHS Strategic Plan 2023–2025](#) (OSBM agency strategic plans). Structured version: [priorities.json](#).

Why NCDHHS

- **Largest realistic opportunity pool.** 947 of the 1,662 grants in the dataset come from HHS. After removing grants that close before 2026-09-30, those that don't list state governments as eligible, and NIH research grants, a quick keyword count finds **roughly 110 DHHS-relevant grants**. The same count finds about 29 for DPI and 11 for DEQ, so a top-10 list for DHHS is a real selection rather than whatever happens to exist.
- **The plan asks for this.** The plan's Medicaid-expansion work is framed as a way to "*maximize federal resources*" (p. 3), and Goal 5, Objective 2.1, is about tracking "new and existing Federal funding." A tool that finds federal money fits a stated priority.
- **The hard part is visible.** DHHS is eligible for almost everything on paper: 609 of the 689 NIH grants list "state governments." The real question is whether DHHS is the *intended* applicant. That makes eligibility reasoning, rather than keyword matching, the core of the problem.

User

The NCDHHS federal-grants coordinator. Each week they review new Grants.gov postings, decide which ones are worth pursuing, and route each one to the division that would own the application (for example Public Health; MH/DD/SUS; Child and Family Well-Being; Medicaid). They are not technical. They need a short list with a reason, an owner and a risk for each grant.

Our "good match" criteria (set before scoring)

1. **Right applicant role.** DHHS can lead, or is an expected partner. This is a gate: confirmed ineligible means excluded. "Technically eligible but designed for university researchers" ranks low.
2. **Strategic alignment.** The funded activity advances a named goal and objective of the 2023–2025 plan (G1–G5).
3. **A division can own it.** Some DHHS division already does this kind of work and has the authority to run it.
4. **Realistic deadline.** For open grants, enough time to apply (for example at least 30 days after the as-of date). Forecasts go to a separate watchlist to prepare early.
5. **Worth the effort.** The award size justifies the application cost; cost sharing is flagged.

Known limitations

- The plan covers 2023–2025. As of September 2026 it may lag the Department's current priorities, especially after federal funding changes in 2025–2026.
- DHHS is very large, so "operational fit" depends on the division, which is why Layers 2 and 3 predict the owning division.
- The division list in `priorities.json` is our summary of the DHHS org chart and should be checked against the current one.

Slide guide: what each slide means, in plain language

For the team: what each slide is saying, why it's there, and what to say if a judge asks. Spoken script: `recording_script.md`.

The big picture in one sentence: we take 1,662 federal grant listings and narrow them to the 15 that NC DHHS should act on. Plain Python handles the facts, two AIs handle the reading, and a fixed formula makes the final call.

1. Title: "Find North Carolina's federal health money first"

Means: who the tool is for (NC DHHS) and what goes in and comes out (1,662 listings in, a short list with reasons out).

2. Our user

Means: we designed for one real person, the DHHS federal-grants coordinator who routes grants to divisions, and we wrote down five rules for a good grant *before* looking at the data: DHHS can apply, it serves a strategic-plan goal, a division already does this work, there's time to apply, and the money is worth the effort.

Why it's there: the rubric's first criterion is whether the criteria are specific to this agency and user.

3. The problem

Means: reading 1,662 listings by hand is impossible, and keyword search is misleading. 609 of 689 NIH research grants list "state governments" as eligible even though they're built for universities, and 559 listings are only announcements of future grants.

4. Step 1: Python rules, no AI

The four step labels at the top of slides 4–7 show where you are and how many grants are left after each step: 1,303 → 283 → 283 scored → top 15.

Means: before any AI, plain code removes only what's objectively dead: 330 already closed, 17 paperwork notices (records that just transfer an existing award), and 12 archived. It also scores the two factual criteria on a fixed 0–10 scale:

- **Deadline:** 0 if it closes within days, 10 if 60+ days away.
- **Award size:** \$100K → 0, \$1M → 5, \$10M+ → 10.

It also cleans fake numbers like "\$999,999,999".

Why: facts shouldn't be left to an AI that might get them wrong.

5. Step 2: Jev screens all 1,303

Means: Jev is a fast, cheap AI that answers multiple-choice questions and says how sure it is. It answers four questions about every grant: is this DHHS's kind of work, which goal, can DHHS apply, and which division. It throws a grant out only when it's at least 85–95% sure. **283 kept, 1,020 set aside:** 997 as off-topic or only weakly related, and 23 because DHHS can't apply (for example, tribes-only grants).

Why: reading everything carefully is expensive, so this screen narrows the pile while erring toward keeping things.

6. Step 3: DeepSeek reads the 283

Means: DeepSeek is a stronger AI. For each of the 283 grants it writes a fit score, whether DHHS can realistically run it, DHHS's role, the owning division, the main risk and a one-line reason. **It must quote the grant word for word**, and Python checks every quote: 282 of 283 were found exactly. A claim it can't back up is downgraded.

Key point: DeepSeek doesn't remove or rank anything. It only explains.

7. Step 4: a published formula decides (bar chart of weights)

Means: the final score out of 100 is a fixed recipe: strategic fit 30%, can DHHS run it 25%, award size 15%, deadline 15%, eligibility certainty 15%. A few rules sit on top: proven-ineligible scores 0, an unproven quote costs 5 points, and no deadline costs 3. The output is three lists:

- **33 open now**
- **116 opening soon** (forecasts)
- **134 research grants** that suit a university partner

Why: anyone can check how a score was made. The AI never picks the winners.

8. Result 1: apply now

Means: the best grant open today is **Title X Family Planning** (98/100). Public Health would own it, DHHS can lead (verified from the grant text), it's due 11 January 2027 with up to \$22M, and federal records show NC DHHS already won \$14.7M under it.

9. The biggest money is opening soon (watchlist table)

Means: the biggest opportunities are **forecasts**, grants announced but not yet open. 4 of the top 5 are programs NC DHHS has already won (about \$275M combined in FY2022–25). Two (Ryan White HIV Part B and Preschool Development B-5) had forecast open dates that have passed, so they may be live now.

10. Why not just keyword search? (comparison table)

Means: we also built the obvious simple method, matching words to the strategic plan, to prove our approach is better. Keyword search ranks Ryan White (a \$187.6M DHHS program) **#670**, and its **#1 pick is a grant only tribes can apply for**. Our screen correctly set that one aside.

11. How confident should you be? (bar chart + three numbers)

Means: four independent checks, strongest first.

1. **Federal spending records (no AI):** grants Jev threw out had a history of paying state agencies only 2% of the time, against 25% for the ones it kept. And 4 of 5 watchlist programs have already paid NC DHHS.
2. **Evidence:** 282 of 283 AI quotes are real.
3. **Two different AIs agree** on DHHS's role 87% of the time.
4. **A small human check** (31 grants) points the same way, but is too small to prove anything.

Bottom line: trust the watchlist most, and double-check the open list beyond Title X.

12. Where the AI was wrong (required by the competition)

Means: three real mistakes and how we caught them.

1. DeepSeek said DHHS would lead 73 university research grants. Jev disagreed, which exposed it, so we added a rule that moves research grants to their own list.
2. DeepSeek once used the single word "states" as proof. Our quote check rejected it.
3. Meta's AI blocked us because DHHS grants mention HIV and overdoses, so we switched to DeepSeek.

13. Three things we think NC DHHS should do (the last slide)

Means: our recommendation, as three actions:

1. **Check Grants.gov now** for Ryan White Part B and Preschool Development B-5, which may already be open.
2. **Apply for Title X** through the Division of Public Health (due 11 January 2027, up to \$22M).
3. **Share the research list** (134 grants) with a university partner, since DHHS can't lead those itself.

One line at the bottom covers what we'd build with more time (a weekly run on live data that emails the coordinator new matches). The competition requires one known limitation and a "with more time" item: the limitation (our small hand-labeled check) is said on slide 11, and the "more time" item is here.

Layer 1 report

Generated by `src/deterministic.py`. All date math uses the fixed as-of date **2026-09-30**. The data was pulled **2026-08-18**, so anything posted or changed after that is not in it.

Rows

- total: 1662
- open track: 1103
- forecast track: 559
- sent on to Layer 2 (not hard-filtered): 1303

Hard filters (rows stay in the file, flagged)

- kept: 1303
- closed_before_as_of_date: 330
- administrative_award_transfer_notice: 17
- archived_before_as_of_date: 12

Open grants: deadlines

- expired (closed before as-of date): 330
- rolling ("accepted anytime"): 66, of which first-come / until funds run out: 9
- no close date in the data: 109 (of which 11 carry a placeholder year-2090+ date)
- deadline_under_7_days: 77
- deadline_under_14_days: 98
- deadline_under_30_days: 142
- deadline_comfortable: 522

Forecast grants (watchlist track)

- forecasted post date already passed: 295
- forecasted close date already passed (stale?): 234
- post date passed but close not: 88 (may already be open, check Grants.gov now)
- no forecasted post date: 86

Funding

- award_ceiling: 718
- unknown: 580
- average_award: 307
- program_total: 57

- rows where a 0 amount was treated as a placeholder: 114
- rows with a sentinel amount (\$999,999,999, 2^31, 9,999 awards) set to missing: 3
- financial_value_score imputed to 5.0 (no amount at all): 580
- award_floor above ceiling: 0

Cost sharing (structured field, text can upgrade Unknown)

- False: 1545
- True: 117
- structured says False but the text states a percentage: 0

Eligibility signals (never used to filter)

- state_governments listed: 887
- broad eligibility (12+ types listed): 694
- applicant types blank: 0
- eligibility text blank: 367
- state grants but text says higher-ed / tribal / individuals only: 9
- applicant slugs not in our known list: none

Research mechanisms (moved to the research-partnerships list in Layer 4)

- research mechanism: 731 (of which still open or forecast: 687)

HHS signals

- assistance listing 93.x: 949
- federal agency is HHS: 947

Data quality

- description under 100 characters: 30 (blank: 0)
- same title + agency under a different id: 16
- open grants whose archive date has passed: 76

Layer 4 inputs

- financial_value_score: mean 4.95, 176 grants score 8 or more
- deadline_score: present for 994 open grants, blank for forecasts and for the 109 with no real close date

Historical-award check (USAspending.gov)

Objective, non-AI check: who actually received grant awards under each grant's assistance listings in **FY2022-FY2025** (USAspending.gov, award types: block, formula and project grants, cooperative agreements). It answers "*is this a program state agencies like DHHS actually win?*". It does **not** measure strategic fit, and an assistance listing covers every funding notice under it, not only this one.

How recipients are classified: by name rules in `src/award_history.py` (state agency / university / tribal / local government / nonprofit-hospital-company), because USAspending's own recipient-type tags are incomplete (for listing 93.917, NC DHHS received \$187.6M by name but only \$91.2M is tagged as state government). Only the top 200 recipients per listing are fetched; `coverage` is the share of dollars those cover. 'State agency history' = NC DHHS received at least \$100K, or state agencies received at least 25% of the dollars.

Grants checked: **1303** (every grant Jev triaged). Distinct assistance listings queried: **257**.

Top 10 open grants

#	Grant	Verdict	NC DHHS	State-agency share	University share	Coverage
1	Title X Family Planning Services Grants	NC DHHS has received awards	\$14.7M	42%	1%	100%
2	Public Health Crisis Response Cooperative Agreement	no awards found FY2022-FY2025	—	0%	0%	0%
3	Home Study and Post-Release Services for Unaccompanied Alien Children	mostly nonprofits, hospitals or companies	—	0%	0%	100%
4	Emerging Infections Network - Research for Preventing, Detecting, and	mostly universities	\$35K	23%	50%	100%
5	NARMS Cooperative Agreement Program to Strengthen Antibiotic Resistanc	mixed recipients (some state agencies)	—	21%	32%	96%
6	Laboratory Flexible Funding Model (LFFM)	mixed recipients (some state agencies)	—	21%	32%	96%
7	US Travelers Health Research, Surveillance, Communication, and Outreac	mostly universities	\$35K	23%	50%	100%
8	Institute of Education Sciences (IES): National Center for Special Edu	mostly universities	—	0%	90%	100%
9	Innovative Traffic Safety Enforcement (ITSE) Grant Program	other state agencies win this	—	35%	3%	100%

10	Authority to Accept Unsolicited Proposals for Research Partnerships No	mostly universities	—	0%	80%	100%
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1 of 10 have awards to NC DHHS (at least \$100K in FY2022-FY2025); **2 of 10** have NC DHHS or substantial state-agency awards; 1 have no awards in FY2022-FY2025; 0 unknown/unclear.

Top 5 watchlist grants

#	Grant	Verdict	NC DHHS	State-agency share	University share	Coverage
1	HIV Care Grant Program - Part B States/Territories Formula and AIDS Dr	NC DHHS has received awards	\$187.6M	76%	0%	100%
2	Maternal, Infant, and Early Childhood Home Visiting Program (MIECHV)	NC DHHS has received awards	\$21.1M	73%	0%	100%
3	Strengthening Services to Prevent the Infectious Disease Consequences	mostly nonprofits, hospitals or companies	—	2%	2%	100%
4	Preschool Development Grant Birth Through Five (PDG B-5) SMART Grant	NC DHHS has received awards	\$36.3M	82%	5%	100%
5	National HIV Behavioral Surveillance	NC DHHS has received awards	\$30.2M	65%	1%	100%

4 of 5 have awards to NC DHHS (at least \$100K in FY2022-FY2025); **4 of 5** have NC DHHS or substantial state-agency awards; 0 have no awards in FY2022-FY2025; 0 unknown/unclear.

Compared with Layer 2 (Jev) routing

- **Deprioritized by Jev:** 1020. Of these, **24** have a history of awards to NC DHHS or substantial state-agency awards (possible false negatives). 6 of them under a listing where NC DHHS itself received awards.
- **Sent to deep review:** 283. Of these, **109** went mostly to universities with no state-agency recipients among the top 200 (probably research-type programs).
- Grants Jev deprioritized vs sent on, share with state-agency history: 2% vs 25%.

Deprioritized grants with state-agency award history (inspect each)

Grant	Jev relevance / role	Verdict	NC DHHS	State share
HIV Care Grant Program - Part B States/Territories Formula and AIDS Drug Assistance Progra	strong / ineligible	NC DHHS has received awards	\$187.6M	76%
NIH, CDC and FDA Small Business Innovation Research Grant (Parent SBIR [R43/R44] Clinical	weak / ineligible	NC DHHS has received awards (spans 41 listings)	\$41.0M	1%
Strengthening Public Health Systems and Services in Indian Country	weak / ineligible	NC DHHS has received awards	\$1.3M	19%

NIOSH Robotics and Intelligent Mining Technology and Workplace Safety Research (U60)	weak / atypical	NC DHHS has received awards	\$579K	3%
Continuation and Expansion of the National Mesothelioma Virtual Bank for Translational Res	weak / atypical	NC DHHS has received awards	\$579K	3%
World Trade Center Health Program Mentored Research Scientist Career Development Award (K0	weak / atypical	NC DHHS has received awards	\$579K	3%
Specialty Crop Multi-State Grant Program 2026	none / atypical	other state agencies win this	—	92%
Fiscal Year (FY) 2026 Funding Opportunity for Indian Tribes and Intertribal Consortia for	weak / ineligible	other state agencies win this	—	87%
RESTORE Act Centers of Excellence Research Grants Program	none / atypical	other state agencies win this	—	80%
RESTORE Act Direct Component - Non-Construction Activities	none / atypical	other state agencies win this	—	80%
RESTORE Act Direct Component – Construction and Real Property Acquisition Activities	none / atypical	other state agencies win this	—	80%
Outdoor Recreation Legacy Partnership Program (ORLP) Recurring Notice 5 Year	none / atypical	other state agencies win this	—	78%
Readiness and Recreation Initiative (RARI) – Recurring 5 Year Notice	none / partner	other state agencies win this	—	78%
F25AS00379 Highlands Conservation Act - Base Funding Round	none / ineligible	other state agencies win this	—	71%
F25AS00332 Highlands Conservation Act – Competitive Funding Round	none / ineligible	other state agencies win this	—	71%
FY2026 ABPP - Battlefield Land Acquisition Grant	none / atypical	other state agencies win this	—	61%
Defense Security Cooperation University - Research Grants	none / atypical	other state agencies win this	—	48%
F26AS00085 Aquatic Invasive Species Interjurisdictional Grants to the Great Lakes States a	none / ineligible	other state agencies win this	—	45%
F26AS00083 Aquatic Invasive Species Grants to Great Lakes States - Fiscal Year 2026 Great	none / ineligible	other state agencies win this	—	45%
F26AS00084 Aquatic Invasive Species Grants to Great Lakes Tribes - Fiscal Year 2026 Great	none / ineligible	other state agencies win this	—	45%
2026 National Urban and Community Forestry Challenge Cost Share Grant Program	weak / atypical	other state agencies win this	—	42%
Fiscal Year (FY) 2022-2026 Advanced Digital Construction Management Systems (ADCMS)	none / ineligible	other state agencies win this	—	39%
F26AS00104 State ANS Management Plan 2026	none / ineligible	other state agencies win this	—	34%
NOI: PROSPECT Program: Providing Opportunities for Specialized Education in Critical Techn	none / atypical	other state agencies win this	—	26%

Caveat: an assistance listing is shared by many funding notices. A listing states win (e.g. a broad CDC or HRSA listing) can still carry a specific notice meant for others, such as a tribal-only or university-only competition. Treat these as leads to check, not proven misses. Our inspection of each is in `docs/failure_modes.md` (F9).

Verdicts across all checked grants

verdict	grants
mostly universities	926
mostly nonprofits, hospitals or companies	106
mixed recipients (some state agencies)	73
NC DHHS has received awards	71
mixed recipients (no state agencies)	39
no awards found FY2022-FY2025	31
other state agencies win this	24
mostly tribal / Native organizations	18
unclear (top recipients cover too little of the money)	8
unknown (API error or non-standard listing)	6
mostly local governments	1

Validation report

Labeled and resolved: **31** grants (0 awaiting adjudication). Label = 'the coordinator should spend time on this'.

> **31 of 31 labels come from a single labeler** (a team member, not a grants expert), so labeler agreement cannot be measured and the labels themselves are uncertain. With this few labels, the intervals below are wide; read them as a first check, not a measurement.

Do humans agree?

- No double-labeled grants yet.
- Disagreements show how ambiguous 'relevant' is; they cap how well any method can score.

Labeled sample by stratum

stratum	labeled	relevant	population
S1_top100	12	6	100
S2_rank101_400	9	2	300
S3_rank401_plus	10	0	903

Layer 2 (Jev) vs keyword baseline

Jev sends **283** grants to deep review. The baseline is given the same budget (its top 283 by TF-IDF similarity). Recall = share of relevant grants that reach Layer 3. Estimates are weighted by stratum size; intervals are 95% stratified bootstrap.

Method	Recall	95% CI	Precision	95% CI
Jev routing	93%	67%–100%	29%	9%–69%
Keyword baseline (same budget)	71%	33%–100%	28%	9%–53%

Layer 2 false negatives (relevant but deprioritized) - inspect every one

- **v041** Expanding global health security through local partnerships in Ethiopia — Jev: weak (p=0.94), role atypical; P(unrelated or weak)=1.00

Layer 3 recommendation vs human label

- 13 labeled grants were deep-analyzed; **recommended_for_review** matches the human label on 69% (unweighted).
- Recommended but labeled not relevant: 1; labeled relevant but not recommended: 3.

Where the AI was wrong (automatic checks, `ai_error_log.csv`)

check	severity	count
layer2_vs_layer3_division	low	59
layer2_vs_layer3_relevance	high	50
layer2_vs_layer3_relevance	medium	11
layer2_vs_layer3_role	medium	91
unverified_eligibility_quote	high	1
unverified_eligibility_quote	medium	1
unverified_strategic_quote	high	1

Examples:

- **unverified_eligibility_quote** — Partnership for Disaster Health Response System: claimed role 'lead' but its quote was not found in the eligibility text ('states'); role downgraded to 'unclear'
- **layer2_vs_layer3_relevance** — Improving Protection against Influenza and Other Respiratory Pathogens: the US Flu VE Network: Jev said 'weak' (p=0.51) but deep model scored alignment 7/10 - a possible Layer 2 false negative
- **layer2_vs_layer3_relevance** — Navigator Emergency Department Diversion Models for Non-Urgent Mental Health Concerns (R34 Clinical Trial Required): Jev said 'weak' (p=0.51) but deep model scored alignment 7/10 - a possible Layer 2 false negative
- **layer2_vs_layer3_relevance** — Development and Testing of Novel Interventions to Improve HIV Prevention, Treatment, and Program Implementation for People Who Use Substances (R34 Clinical Trial Required): Jev said 'weak' (p=0.5) but deep model scored alignment 7/10 - a possible Layer 2 false negative
- **layer2_vs_layer3_relevance** — Notice of Intent to Publish a Forecast for EmbraceHealth Clinical Research Network to Reduce Health Disparities – Research Sites: Jev said 'weak' (p=0.63) but deep model scored alignment 9/10 - a possible Layer 2 false negative

Failure log

Every failure we find goes here, in this format. Entries F1–F5 were found while building the data layers, before any model run. **Add the model failures (from `data/results/ai_error_log.csv`, the spot-check and validation_report.md) after the real run.** The submission needs at least one place where *the AI* was wrong and how we caught it.

Template:

ID / Grant ID:
Observed error:
Why the system made the error:
How we detected it:
Whether we changed the pipeline:
What tradeoff the fix introduced:

F1. Keyword matching recommends a grant DHHS cannot apply for

- **Grant:** Tribal Maternal, Infant, and Early Childhood Home Visiting Program (`3f13ec1b-3d98-484f-ac0b-6d8169f62831`)
- **Observed error:** The TF-IDF baseline ranks it **#6 of 1,320**, because its wording matches DHHS's home-visiting and child-and-family goals almost perfectly.
- **Why:** The eligibility text says applicants are "federally recognized Indian tribes (or consortia of tribes), tribal organizations, and urban Indian organizations". A state agency cannot apply. Word overlap can't read eligibility.
- **How detected:** Manual review of the baseline's top 10 (`data/results/keyword_baseline.csv`).
- **Pipeline change:** This is why Layer 2 asks Jev for `applicant_role` and Layer 3 must quote the eligibility text. The keyword method stays only as a baseline to compare against.
- **Tradeoff:** Eligibility now depends on a model's reading of the text, which can itself be wrong. That's why the eligibility quote is checked verbatim, and why only *verified* ineligibility zeroes a score.

F2. Structured eligibility says "state governments" on grants not meant for states

- **Observed error:** 609 of the 689 NIH grants list `state_governments` as an eligible applicant type. Filtering on that field would keep hundreds of university research grants, and dropping grants without it would lose any grant that describes state eligibility only in its text.
- **How detected:** Counting applicant types against the funding agency during agency selection.
- **Pipeline change:** Structured eligibility is only a signal (Layer 1 flags). The role decision (lead / partner / atypical / ineligible) comes from reading the text in Layers 2–3.
- **Tradeoff:** More grants reach the paid model layers (1,320 instead of a few hundred). The cost is still under \$1.

F3. Placeholder award amounts would have caused large-dollar bias

- **Observed error:** One grant lists an award ceiling of **\$999,999,999**, and another lists total funding of **\$2,147,483,647** (2^{31} , a software maximum). Read as money, they would get the maximum financial score.
- **How detected:** Range checks on the award fields. Layer 1 now treats 3 sentinel rows and 114 zero amounts as placeholders (docs/layer1_report.md).
- **Pipeline change:** Placeholders and zeros are set to missing. Missing amounts get a neutral fixed score (5/10).
- **Tradeoff:** A real grant that omits its amount is slightly under-ranked on award value.

F4. "Forecasted" does not mean "upcoming"

- **Observed error:** 234 of the 559 forecasts have a forecasted close date **before** our as-of date (2026-09-30). They were probably already posted under another record, delayed, or dropped.
- **How detected:** Layer 1 date check.
- **Pipeline change:** Kept on the watchlist, since the program may still come back, but flagged as `stale_forecast` and shown as "date passed — check if posted".
- **Tradeoff:** The watchlist may contain some programs that no longer exist. We chose that over silently dropping real ones.

F5. The strategic plan is out of date

- **Observed error:** The only NCDHHS plan on the OSBM page covers **2023–2025**. Our alignment scores measure fit with that plan, not necessarily with 2026 priorities (for example after federal funding changes).
- **How detected:** Reading the plan's date range.
- **Pipeline change:** None possible with the provided data. Stated as a known limitation.
- **Tradeoff:** Not applicable. Leadership should read "strategic alignment" as alignment with the 2023–2025 plan.

F6. A commercial AI provider blocked legitimate public-health work

- **Observed error:** Partway through Layer 3, Meta's provider (Muse Spark 1.3 contributor) **restricted our account for "repeated policy violations"**. Every later request failed, including a test that only said "Reply with the word OK".
- **Why:** Our only inputs were public Grants.gov announcements and the DHHS strategic plan. DHHS grants routinely discuss HIV, STIs, sexual violence, overdoses and child abuse, and the provider's safety filter evidently counted that as violations. 62 of 298 grants had already been analyzed.
- **How detected:** Layer 3 progress stalled, and a direct test request returned HTTP 403 with the provider's message.
- **Pipeline change:** We tested three replacement models on the 8 most sensitive grants Muse had analyzed (valid output, verified quotes, agreement with Muse, cost) and switched to **DeepSeek V4 Pro**: 8/8 valid, 8/8 quotes verified, 100% agreement with Muse on applicant role and division. The switch was one setting, because the model is configurable and every answer is cached.

- **Tradeoff:** Higher cost (~\$2.50 instead of ~\$0.30 for 298 grants) for a more capable model that doesn't refuse this subject matter. **For a real DHHS deployment this is a vendor risk:** the agency's normal vocabulary can trip a vendor's safety filter, so the system must not depend on one AI provider.

F7. The in-depth model said DHHS would "lead" NIH research grants

- **Observed error:** DeepSeek V4 Pro (Layer 3) labeled **73 research or training grants** (NIH R01/R34/R61, CDC/FDA U01 ...) as ones DHHS would *lead*, because "state governments" appears on their eligibility lists. With full eligibility credit, **7 of the first top-10 "apply now" grants were NIH research awards**, which a state grants coordinator can't realistically pursue as lead applicant.
- **Why:** The Layer 3 prompt lists the role options but, unlike the Jev prompt, never defines them. "Lead" was read as "legally allowed to apply".
- **How detected:** (1) Reading the top 10, where the model's own "main risk" text said "requires an academic partner". (2) The Layer 2 vs Layer 3 disagreement log: Jev labeled **0** of those grants as DHHS-led (65 partner, 61 atypical, 4 ineligible).
- **Pipeline change:** An objective Layer 1 rule, `research_mechanism` (NIH opportunity, or a research/training activity code such as R01, K99, U01 or P50 in the title; deliberately not CDC's U60 state cooperative agreements). Layer 4 moves these 134 grants to a separate **research-partnerships** list for university partners.
- **Tradeoff:** A rare research award DHHS *could* lead now sits on the research list instead of the apply-now list. The prompt itself still lacks role definitions; fixing it means re-running Layer 3 (about \$3).

F8. Administrative paperwork listed as funding opportunities

- **Observed error:** 17 records are not opportunities at all: 15 CDC placeholders titled only "RFA-XX-18-000" ("Submit application as necessary [for Type 6 Applications]", close date 2030), plus NIH successor-in-interest / change-of-recipient notices. They exist only to transfer an existing award. Several reached the top 10, because the AI rated a vague "public health" placeholder as a plausible fit.
- **How detected:** Reading the top 10 (entries titled "RFA-TS-18-000", "RFA-OE-18-000"). The AI's own risk note said "the announcement provides almost no programmatic detail". These records also caused most of the unverified quotes, since there's nothing real to quote.
- **Pipeline change:** Layer 1 hard-filters them (`administrative_award_transfer_notice`). The test pins the count at 17.
- **Tradeoff:** None for a grants coordinator; nobody can win new money through these notices.

F9. Two independent checks disagreed: Jev vs. the historical-award check (Jev was right)

- **Observed:** The USAspending check (`data/results/award_history_report.md`) flagged **24** grants Jev set aside as having "state-agency award history", which made them possible false negatives. The most striking one is **"HIV Care Grant Program - Part B ... Pacific Islands Jurisdictions" (HRSA-27-064)**: Jev rated it *strong* relevance but *ineligible* (p = 0.94), while NC DHHS received **\$187.6M** under its assistance listing (93.917) in FY2022–25.

- **Why they disagree:** An assistance listing is shared by every funding notice under it. HRSA-27-064 is the Pacific-territories version of Ryan White Part B; its eligibility text reads "Domestic territories, and freely associated states". NC's own version, HRSA-27-063, is #1 on our watchlist. **Jev read the notice correctly; the listing-level history could not tell the two notices apart.**
- **The other 23, from titles and eligibility:** the NIH/CDC/FDA small-business (SBIR) parent notice, which spans 41 listings; two tribal-only programs; three research programs (NIOSH mining robotics, a mesothelioma tissue bank, World Trade Center research careers); and 17 programs outside DHHS's field that states do win (Gulf RESTORE Act, conservation, outdoor recreation, invasive species, urban forestry, specialty crops, battlefield land, defense research, construction technology, specialized education). **None looked like a genuine DHHS miss.**
- **How detected:** Building the historical-award check and comparing it against Layer 2's routing, then reading the eligibility text of each flagged grant.
- **Pipeline change:** None to the ranking. The report carries the caveat that history is per listing, not per notice, and marks grants spanning more than 3 listings.
- **Tradeoff / lesson:** The award history is a useful *objective* signal: grants Jev set aside had state-agency history **2%** of the time, against **25%** for grants it sent on. But it can't replace reading the specific notice.

F10. Most of the "apply now" list isn't money DHHS has won before

- **Observed:** Of our **top 10 open** grants, only **1** (Title X, \$14.7M to NC DHHS in FY2022–25) has a history of awards to NC DHHS. #2 (Public Health Crisis Response) has no awards at all in FY2022–25: it's a new roster mechanism that pays only during emergencies. Of the **top 5 watchlist** grants, **4** do: Ryan White Part B \$187.6M, Preschool Development B-5 \$36.3M, HIV Behavioral Surveillance \$30.2M, MIECHV \$21.1M.
- **Why:** Most of the state programs DHHS traditionally wins either closed between the August 18 data pull and September 30, or are forecasts. The open list is what's left.
- **How detected:** The historical-award check (USAspending.gov), an objective source that's independent of both AI models.
- **Pipeline change:** None yet. With more time, award history could become a scoring input (for example, a bonus for programs states actually win). It's deliberately kept as validation here, so the check stays independent of the ranking it checks.

Label uncertainty (validation)

- The first 31 labels come from **one team member, not a grants expert** (`data/validation/labels.csv`, labeler A). Jev's only "miss" among them, "*Expanding global health security through local partnerships in Ethiopia*" (labeled relevant), is arguably a labeling call rather than a model error: it's an overseas CDC program. We kept the label as given. Recall numbers from 8 relevant grants have wide intervals and should be read as a first check.

More model failures (add from the labeled sample once it is done)

Sources: `data/results/ai_error_log.csv` (unverified quotes, invalid outputs, Layer 2 vs Layer 3 disagreements), `data/validation/spotcheck.csv`, and the false-negative list in `data/results/validation_report.md`.